

JEE MAIN REVISION MASTER: WAVES SYLLABUS REDUCED

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 96 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Waves - Core Application	96 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
-----------------	--

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Waves - Core Application (Very High Yield — 96 PYQs)

Core Principle: Key problem pattern in Waves.

■ ■ Caution / JEE Pitfall: Watch for algebraic and sign errors.

■ Speed Trick: Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Morning Shift] Q#1 (MCQ): "The equation of a plane progressive wave is given by $y=5\cos\pi(200t-(x/150))$ where x and y are in cm and t is in second. The $v...$ " [Ans: D]
- [2026 Morning Shift] Q#1 (MCQ): "Two strings (A,B) having linear densities $\mu_-(A)=2 \times 10^{-4}\text{kg/m}$ and, $\mu_-(B)=4 \times 10^{-4}\text{kg/m}$ and lengths $L_-(A)=2.5\text{m}$ and ..." [Ans: B]
- [2026 Evening Shift] Q#2 (MCQ): "The fifth harmonic of a closed organ pipe is found to be in unison with the first harmonic of an open pipe. The ratio of lengths of closed p..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 2nd April Morning Shift (Q#1)

The equation of a plane progressive wave is given by $y=5\cos\pi(200t-(x/150))$ where x and y are in cm and t is in second.

The velocity of the wave is _____ m/s.

- (A) 120
- (B) 150
- (C) 200
- (D) 300 [Correct]

Correct Answer: D

Solution Analysis: A plane progressive wave moving in the positive x -direction can be represented as: $y=A\cos(\omega t-kx)$ Where, ω is the angular frequency, k is the propagation constant (wave number). The given equation is $y=5\cos\pi(200t-(x/150))$. $y=5\cos(200\pi t-(\pi x/150))$ By comparing this with $y=A\cos(\omega t-kx)$, $\omega=200\pi$ rad/s and $k=(\pi/150)$ rad/cm Wave velocity is the distance travelled by the wave per unit time. It is related to angular frequency and wave number by the relation: $v=(\omega/k)$ Substituting the values of ω and k : $v=\frac{200\pi}{\pi/150}$